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CHE 313 Transport Phenomena III

Exam #1 (2/19/2020)

Name: Rayan Bae

We are separating acetone and ethanol in a distillation column with a total condenser and a partial reboiler. The column has two feeds and operates at a pressure of 1.0 atm. Feed 1 is a saturated liquid that is 60.0 mol% acetone and has a feed rate of $F_1 = 200.0$ kmol/hr. The second feed is 40.0 mol% acetone, has a feed rate of $F_2 = 300.0$ kmol/hr, and is a two-phase mixture that is 20% vapor. Assume CMO is valid. Bottoms product is 10.0 mol% acetone, and distillate product is 80.0 mol% acetone. External reflux ratio $L/D = 2.0$.

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- 1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables (use V & L for enriching section, V' & L' for middle section, and $\bar{L}$ & $\bar{V}$ for stripping section)

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- 2) (10 pts) Draw mass balance envelopes for each section (total four balance envelopes)

Page 2

- 3) (15 pts) Find D , B , L , V , V' , L' , $\bar{L}$, and $\bar{V}$ (hint: set up overall mass balance and component mass balance)

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- 4) (5 pts) Find slope, y-intercept, and coordinate on $y=x$ curve of the top operating line:

$$y = \left(\frac{L}{V}\right)x + \left(1 - \frac{L}{V}\right)x_D$$

- 5) (5 pts) Find slope, x-intercept, and coordinate on $y=x$ curve of the bottom operating line:

$$y = \left(\frac{\bar{L}}{\bar{V}}\right)x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$$

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- 6) (10 pts) The equation for the feed line is:

$$y = \frac{q_i}{q_i - 1}x + \frac{z_i}{1 - q_i}$$

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$$q_i = \frac{L_{Fi}}{F_i}$$

Find q_1 , q_2 , coordinates on $y=x$ line, and slopes of two feed lines (F_1 & F_2), and x-intercept of F_2 .

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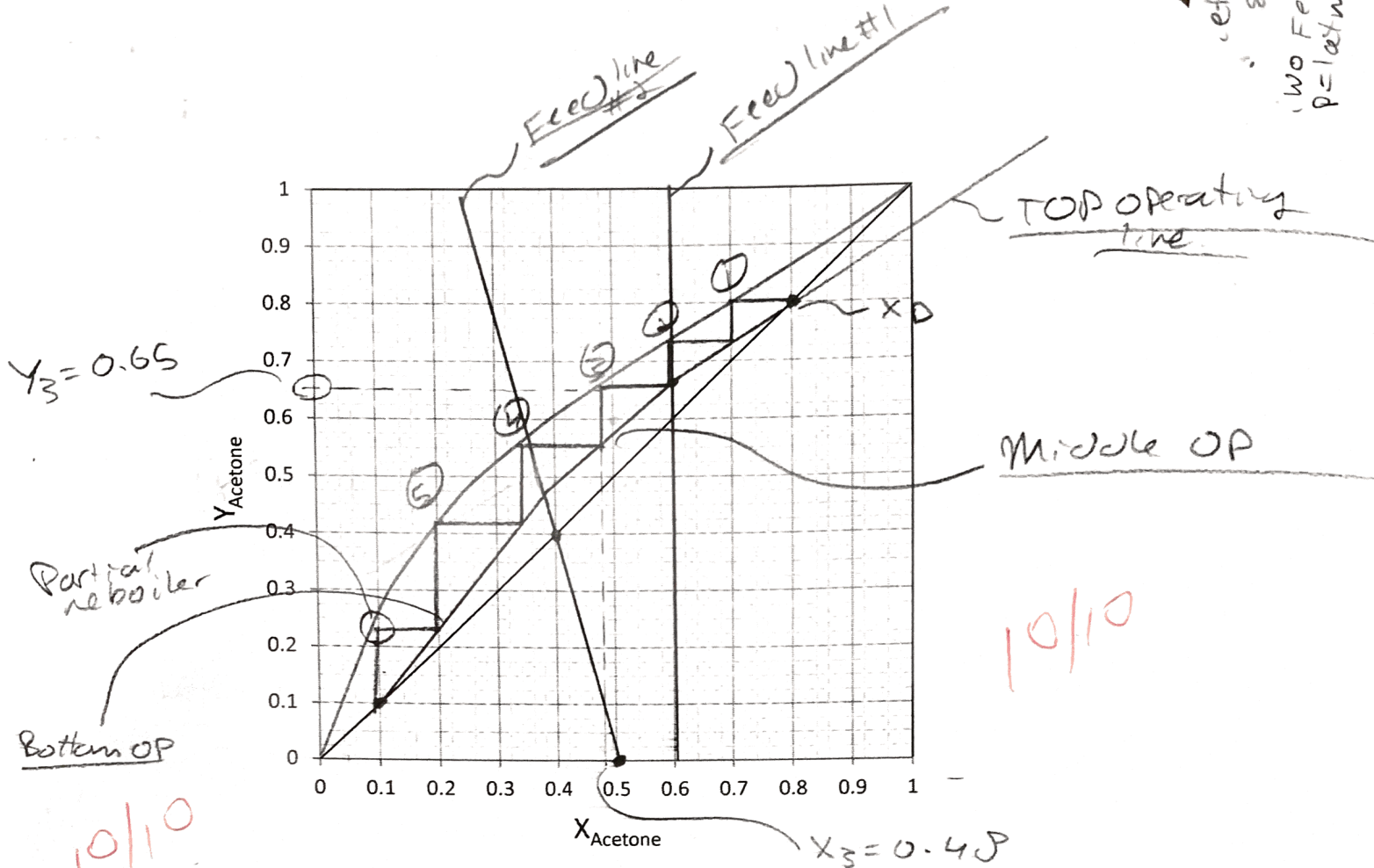
- 7) (5 pts) Write overall mass balance and component mass balance around the top column (hint: mass balance between top feed and distillate) $\rightarrow$ OP for middle section

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- 8) (15 pts) Derive an operating equation for the middle section. Find slope and y-intercept

- 9) (10 pts) Draw the top operating, bottom operating, two feeds, and middle operating lines

Back side



10) (10 pts) Find the number of stages and the optimum locations of feed 1 and feed 2 stages

11) (10 pts) Find mole fraction (x_3 , y_3) of the stage 3

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✓ TOTAL STAGES = 5 + Partial reboiler

optimum stage of
feed #1 feed #2
Stage 2 Stage 4

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STAGE #3

$x_3 = 0.48$

$y_3 = 0.65$

✓

ethanol w/ distillation
total condenser & partial reboiler

Ryan
Baig
Page 1

Two Feeds
 $P = 1 \text{ atm}$

Feed 1

Sat. Liquid

$z_1 = 0.60$

$F_1 = 200.0 \frac{\text{kmol}}{\text{hr}}$

Feed 2

Two phase

$z_2 = 0.40$

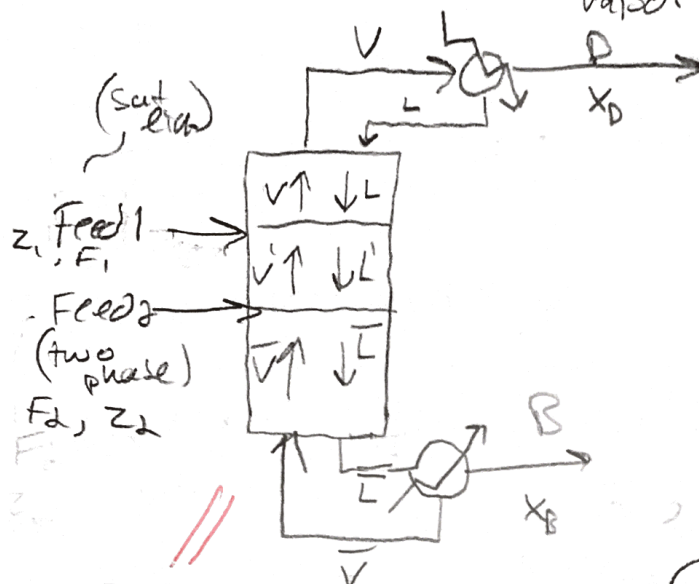
$F_2 = 300.0 \frac{\text{kmol}}{\text{hr}}$

vapor is 60%

$x_B = 0.10$

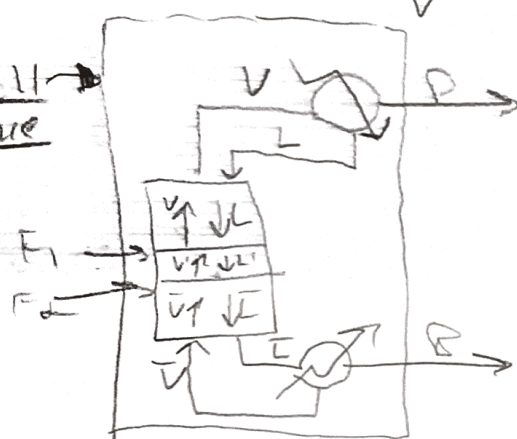
$x_D = 0.80$

$\frac{L}{D} = 2.0$



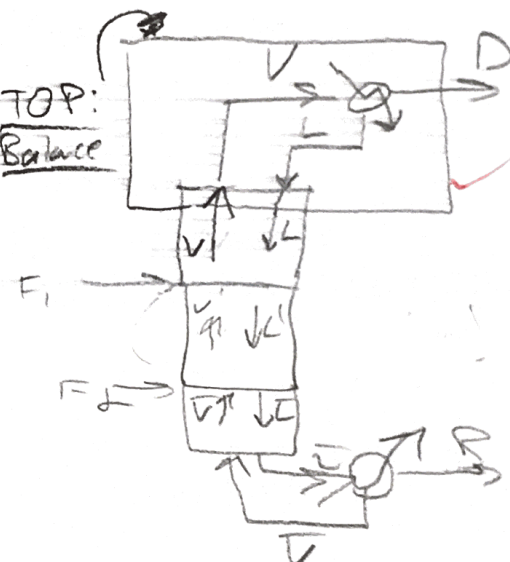
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2) Overall Balance

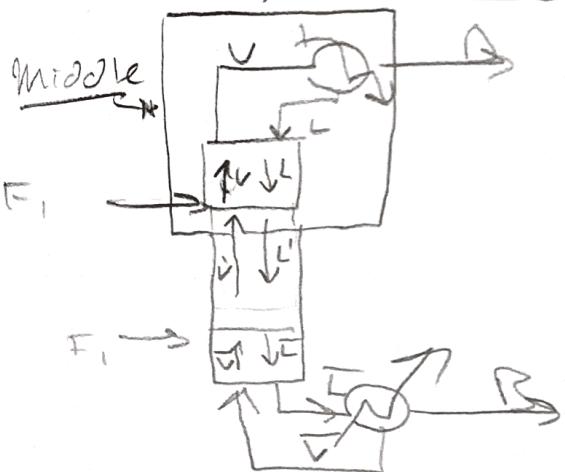


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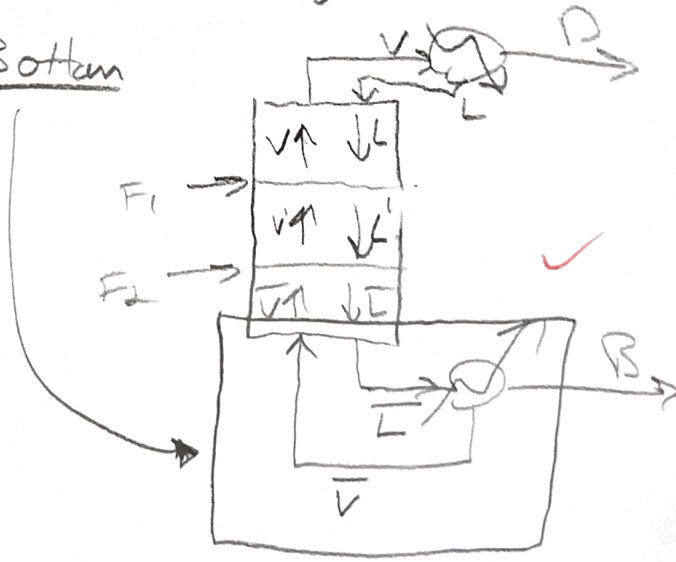
TOP: Balance



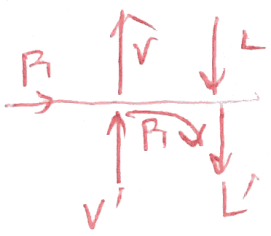
Middle



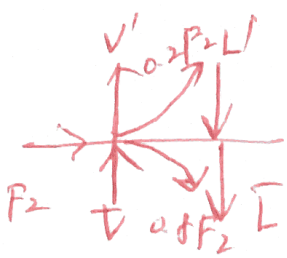
Bottom



Feed 1



Feed 2



$$F_1 + F_2 = D + B$$

$$F_1 z_1 + F_2 z_2 = D x_D + B x_B$$

Two unknowns & 2 eqs.

$$(200)(0.60) + 300(0.4) = D(0.8) + B(0.10)$$

$$D = 200 + 300 - B$$

$$(200)(0.60) + 300(0.4) = (500 - B)(0.8) + B(0.10)$$

$$B = 228.571 \frac{\text{kmol}}{\text{h}}$$

$$D = 271.429 \frac{\text{kmol}}{\text{h}}$$

$$F_1 = 200$$

$$z_1 = 0.60$$

$$F_2 = 300$$

$$z_2 = 0.40$$

$$x_D = 0.80$$

$$x_B = 0.10$$

Raym
Balg
pased

assume
CMO
valid!
*

15/15

$$w \quad \frac{L}{D} = 2.0 \Rightarrow$$

$$L = 542.857 \frac{\text{kmol}}{\text{h}}$$

MB of TOP

$$V = L + D \Rightarrow$$

$$V = 814.286 \frac{\text{kmol}}{\text{h}}$$

$$V' = 814.286 \frac{\text{kmol}}{\text{h}}$$

$$V = V'$$

$$V' = F_2(0.20) + \bar{V}$$

20% vapor of feed 2

$$\Rightarrow V = F_2(0.20) + \bar{V}$$

$$\bar{V} = V - F_2(0.20)$$

$$\bar{V} = 814.286 - (300)(0.20)$$

$$\bar{V} = 754.286 \frac{\text{kmol}}{\text{h}}$$

Bottom MB

$$\bar{L} = B + \bar{V} = 228.571 + 754.286$$

$$\bar{L} = 982.857 \frac{\text{kmol}}{\text{h}}$$

$$L' = F_1 + \bar{L} = 200 + 982.857$$

sat liquid

$$L' = 1182.857 \frac{\text{kmol}}{\text{h}}$$

4) TOPOP

$$\text{slope} = \frac{L}{V} = \frac{2}{3}$$

$$y\text{-intercept} \quad \left(1 - \frac{2}{3}\right) \frac{8}{10} = \frac{L}{15}$$

$$y\text{-intercept} = \frac{4}{15}$$

$$y=x \text{ coordinate is @ } y=x=0.80$$

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$$\text{Slope} = \frac{L}{V} = \frac{982.857}{754.286} = \boxed{1.303} \quad \checkmark$$

X-intercept

→ @ y=0
find x

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$$y = \left(\frac{982.857}{754.286} \right) x - \left(\frac{982.857}{754.286} - 1 \right) (0.10)$$

$$\left(\frac{982.857}{754.286} - 1 \right) (0.10) = x \left(\frac{982.857}{754.286} \right)$$

$$\boxed{\text{X-intercept} = 0.0233} \quad \checkmark$$

$$\boxed{\text{Coordinate on } y=x = x_B = 0.10} \quad \checkmark$$

6) Feedline 10/10

$$q_1 = \frac{L_{\text{below}} - L_{\text{above}}}{F_1} = \frac{L' - L}{F_1}$$

$$\boxed{q_1 = 1} \quad \checkmark$$

$$L' = F_1 + L$$

$$\left(\frac{F_1 + L}{F_1} \right) - \frac{L}{F_1} = 1$$

$$q_2 = \frac{L_{\text{below}} - L_{\text{above}}}{F_2} = \frac{L - L'}{F_2}$$

$$\boxed{q_2 = 0.8} \quad \checkmark$$

$$\begin{aligned} L &= F_2(0.8) + L' \\ \frac{F_2(0.8) + L' - L'}{F_2} &= 0.8 \end{aligned}$$

coordinates on y=x line → FOR FEED #1 → $\boxed{y=x=0.60}$

→ FOR FEED #2 → $\boxed{y=x=0.40}$

Slope of feedline #1

$$\text{Slope} = \frac{q}{q-1} = \frac{1}{1-1} = \frac{1}{0} = \boxed{\infty} \quad \checkmark$$

Slope of feedline #2

$$\text{Slope} = \frac{q}{q-1} = \frac{0.8}{0.8-1} = \frac{0.8}{-0.2} = \boxed{-4} \quad \checkmark$$

X-intercept of F_2

$$y = -4x - \frac{2}{q-1} = ?$$

$$y = -4x - \frac{0.4}{0.8-1} \Rightarrow$$

$$y = -4x + 2$$

$$0 = -4x + 2 \Rightarrow \frac{-2}{-4} = x$$

$$\boxed{\text{X-intercept of } F_2 = \frac{1}{2}} \quad \checkmark$$

all MBS TOP

$$V = D + L$$

Component MBS TOP

$$V_y = D x_D + L x$$

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Page #4

8) operating eq. for middle sec

MB: $F_1 + V' = L' + D$

CB: $F_1 z_1 + V' y = L' x + D x_D$

$$V' y = L' x + D x_D - F_1 z_1$$

$$y = \frac{L'}{V'} x + \left(\frac{D x_D - F_1 z_1}{V'} \right)$$

(7) MB bet. top feed & distillate!

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$$\text{Slope} = \frac{L'}{V'} = \frac{742.857}{814.286} = 0.912$$

$$\text{slope} = 0.912$$

$$y_{\text{intercept}} = \frac{D x_D - F_1 z_1}{V'} = \frac{(271.429)(0.80) - (200)(0.60)}{814.286}$$

$$y_{\text{intercept}} = 0.119$$

Summary of all lines

Feed 1: Slope = ∞ intersects $y=x=0.60 \therefore x=0.60$

Feed #2: $x_{\text{intercept}} = 0.5$ & $y=x=0.$

TOP OP:

$$y = \frac{2}{3} x + \frac{4}{15} \Rightarrow \Rightarrow \text{uses } x_D$$

$$\text{e } x=0.5 \text{ @ } x=0.5 \quad y=0.6$$

Bottom OP

intersects w/ Feed 2 & $x_B = 0.10$

$$y = 1.303 \rightarrow x_{\text{int}} = 0.023$$

Middle

$$y = 0.912 x + 0.119$$